

Aerodynamic Optimization of a Cybertruck Geometry: Turbulence Model Validation and Rear Wing Design

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I. INTRODUCTION

The automotive industry is under increasing pressure to improve vehicle aerodynamics to increase the energy efficiency and driving range of electric vehicles. Wind tunnel testing, a key component in aerodynamic optimisation, is expensive and thus limits the number of design iterations that can be performed in the early stages of development. Computational Fluid Dynamics (CFD) is a promising alternative for fast, drastically reduced cost assessment of design concepts. But the reliability of CFD predictions is highly dependent on correct model set-up, turbulence model choice and proper validation with experimental data.

The current study focuses on the aerodynamic performance of a simplified Cybertruck geometry using steady-state RANS simulations. Two turbulence models are considered: the standard $k - \epsilon$ (SKE) model and the Shear Stress Transport (SST) model. The main goal is to evaluate the validity and credibility of these numerical models through systematic verification and validation processes, thus choosing the most appropriate model for aerodynamic design optimization.

The work is divided into two parts. In Part I, development of a robust CFD methodology through sensitivity analyses on the mesh and validation with reference data, including the comparison of drag and lift coefficients, surface pressure distributions and flow field characteristics, is presented. Part II discusses the practical application of CFD to design exploration by examining the aerodynamic effect of a particular design change. The Reynolds number based on the square root of frontal area is 588,000, which corresponds to realistic highway driving conditions at 5 m/s. The aim of this work is to demonstrate that CFD

can be a useful tool in the early stages of aerodynamic design by developing a modelling approach that is both computationally efficient and accurate. Although expensive physical testing will be required to validate the design, the modelling approach will be predictive enough to allow design decisions to be made.

II. METHOD

Computational Domain and Boundary Condition The computational domain was a rectangular box of $10L \times 2L \times 2.5L$ in the streamwise (x), lateral (y) and vertical (z) directions, respectively, where L is the truck length. The vehicle geometry was positioned with its front face $3.5L$ downstream of the inlet and its rear face $6.5L$ upstream of the outlet to provide an adequate distance for flow development and wake evolution and to minimise boundary effects. A symmetry plane was enforced over the vehicle centerline to reduce the computational costs since the computational domain is half of the physical geometry.

The inlet boundary condition was set to a uniform velocity of 5 m/s in the streamwise direction. Inlet turbulence conditions were specified using a turbulence intensity of 1% and a turbulence length scale of 0.1 m. The turbulence intensity represents the ratio of root-mean-square velocity fluctuations to the mean flow velocity and was selected based on typical values for low-turbulence external flows (1-5%) as recommended by the ANSYS Fluent Theory Guide [1].

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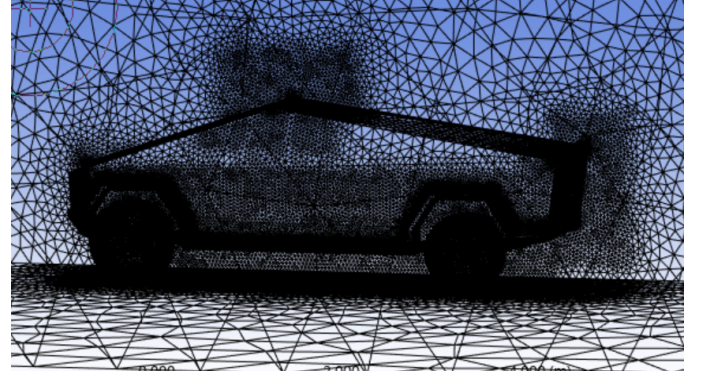
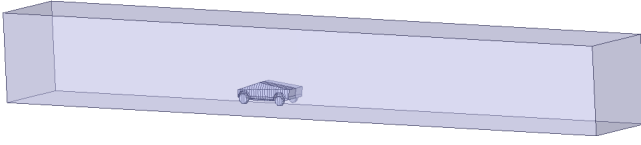


Fig. 1. Computational domain and mesh configuration for truck aerodynamic simulation, showing (left) the full domain extent with inlet, outlet, and symmetry boundaries, and (right) detailed surface mesh on the vehicle geometry with refined boundary layer resolution and localized refinement in the near-wall and wake regions

The outlet was defined as a pressure outlet with zero gauge pressure. The surface of the vehicle was modelled as a no-slip smooth wall and the ground as a moving wall translating at the free-stream velocity (5 m/sec) to simulate realistic road conditions, and to remove the artificial boundary layer that would form on a stationary ground. The symmetry conditions were applied to the side and top boundaries of the domain as they were sufficiently far from the vehicle (1L and 1.25L, respectively) to avoid any blockage effects.

Mesh Generation and Verification

Mesheres were generated for the SKE and SST turbulence models, according to their respective near wall resolution. Both meshes were unstructured tetrahedral/hexahedral hybrid meshes with inflation layers extruded off all vehicle surfaces to capture the boundary layer. For the SKE model ten inflation layers with a growth ratio of 1.2 and a first layer height of 4.43×10^{-3} m resulted in a target y^+ of around 30. A body of influence with an element size of 0.05 m was applied around the vehicle to capture the wake structures and flow features leading to a mesh of the order of 900,000 nodes. For the SST model, a target y^+ of 1 was achieved with twenty inflation layers with a growth ratio of 1.2 and a first layer height of 1.45×10^{-4} m. A mesh sensitivity study was conducted with a coarser mesh with a refinement factor of 1.41 applied to the bulk mesh element sizes. The near-wall mesh parameters (first layer height and number of layers) were kept constant to meet turbulence model requirements. The mesh independence was verified by comparing the drag coefficient (CD) obtained from fine and coarse meshes and less than 3% variation was observed, which confirmed the adequate mesh resolution for both turbulence models.

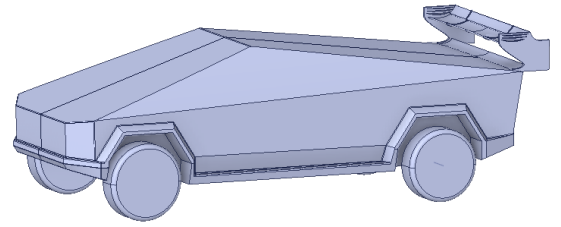


Fig. 2. Modified truck geometry with wing for Part II aerodynamic analysis

III. RESULTS & DISCUSSION

Drag and Lift Coefficients

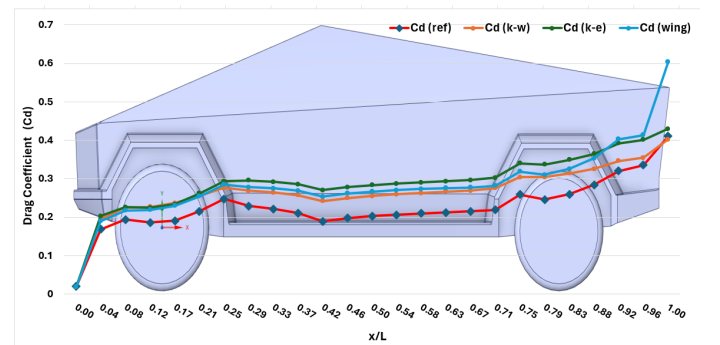


Fig. 3. Drag coefficient distribution for different models along the length of the car

Both the standard $k - \epsilon$ (SKE) and $k - \omega$ SST turbulence models were in reasonable agreement with the reference data in terms of the overall drag accumulation along the truck length. In the front and mid-sections ($x/L = 0-0.8$) both models predicted accurately the drag buildup with coefficients that remained relatively constant

around $C_D \approx 0.20-0.30$. However, large deviations appeared in the rear region ($x/L > 0.85$), where both models systematically overpredicted the drag. The final drag coefficient from the $k-\omega$ SST model was $C_D \approx 0.40$ (14% overprediction of reference $C_D \approx 0.35$). The $k-\epsilon$ model had larger overprediction at $C_D \approx 0.42$ (20% error). This discrepancy in the rear is due to the inherent limitations of steady-state RANS modelling in resolving the complex three-dimensional vortex structures and massive flow separations typical of bluff body wakes. The eddy viscosity of the $k-\epsilon$ model is overestimated in separated regions, and the turbulent mixing is enhanced, resulting in larger wake zones and drag. The $k-\omega$ SST model performs better because it is a blended model (uses $k-\omega$ near the wall for proper prediction of boundary layers and separation and gradually switches to $k-\epsilon$ in the freestream) and thus it represents the flow physics more balanced in areas with adverse pressure gradients.

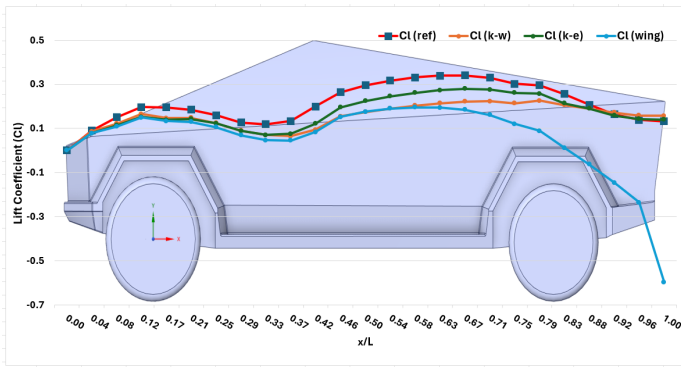


Fig. 4. Lift coefficient distribution for different models along the length of the car

The lift coefficient predictions indicated that the two turbulence models and the reference data exhibited comparable patterns of agreement and divergence. In the front and early mid-section regions ($x/L=0-0.5$), both $k-\epsilon$ and $k-\omega$ SST models were in excellent agreement with reference values. The gradual lift build-up as flow accelerates over and under the truck body was captured accurately with lift coefficients of $C_L \approx 0.10$ to 0.20 . However, in the critical mid-to-rear section ($x/L = 0.5-0.9$), where the slanted roof geometry creates a strong low pressure region on the upper surface, both models significantly underpredicted lift as compared to the reference data. The reference data showed a lift coefficient peak in this region of about $C_L \approx 0.35-0.40$, the $k-\omega$ SST model predicted $C_L \approx 0.20-0.25$ (30-40% underprediction) and the $k-\epsilon$ model predicted $C_L \approx 0.22-0.28$ (25-35% underprediction). This systematic underprediction is due to the inability of the RANS models to accurately resolve

the magnitude of the suction peak on the slanted surface, likely due to the overprediction of the eddy viscosity that dampens velocity gradients and reduces the pressure differential between upper and lower surfaces. In addition, the models are poor at predicting the size and location of the separation bubble on the slant. The models' inability to capture unsteady vortex shedding phenomena contributes to the underprediction of lift. Although these limitations are well-known difficulties for steady RANS approaches in separated flows, the $k-\omega$ SST model displayed consistently better results in both drag and lift predictions. Its better near-wall treatment, better treatment of adverse pressure gradients and better prediction of flow separation make it the obvious choice for the design investigation. The results of the $k-\omega$ SST model were with 10-15% error for drag and 20-30% for lift, which is acceptable in the preliminary aerodynamic design studies using steady-state RANS simulations.

Surface and Flow Parameters

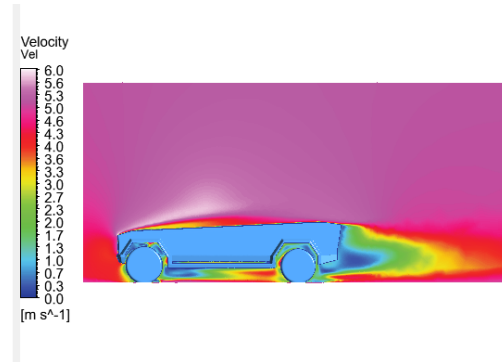


Fig. 5. Velocity magnitude contour for $k-\epsilon$ turbulence model on the plane at $z = 0.96$ m

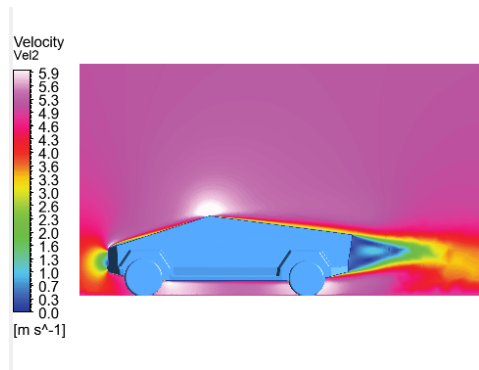


Fig. 6. Velocity magnitude contour for $k-\epsilon$ turbulence model on the centerline plane

Figures 5,6,7 and 8 display the distribution of the velocity magnitude for both the $k-\epsilon$ and $k-\omega$ SST turbulence models. Both models predict similar flow features around

the truck geometry, such as a frontal stagnation zone with near-zero velocity, acceleration of the flow over the roof and bonnet surfaces beyond the freestream conditions and localised acceleration in the underbody and wheel well regions. In both cases, the wake structure behind the blunt rear surface shows a large velocity deficit, indicating flow separation and recirculation behaviour typical of square-back vehicle configurations.

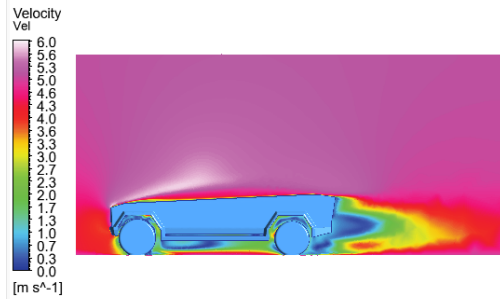


Fig. 7. Velocity magnitude contour for $k - \omega$ turbulence model on the plane at $z = 0.96$ m

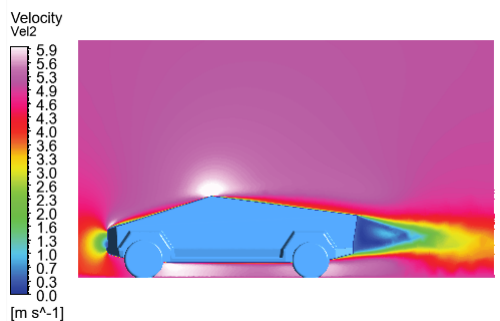


Fig. 8. Velocity magnitude contour for $k - \omega$ turbulence model on the centerline plane

The comparison shows some small but important differences between the two turbulence formulations. The $k - \omega$ SST model produces a slightly narrower wake structure and improved velocity gradients at the wake boundaries, indicating improved resolution of the development of the shear layer and earlier momentum recovery downstream of the separation point. The underbody flow visualisation indicates a slightly stronger acceleration in localised regions for the $k - \omega$ model due to the better near-wall treatment and prediction of the boundary layer.

The $k - \omega$ SST model is generally expected to provide a better agreement with the experimental data for separated flows. Fig. 9 and Fig. 10. show the distribution of the skin friction coefficient on the truck surface for

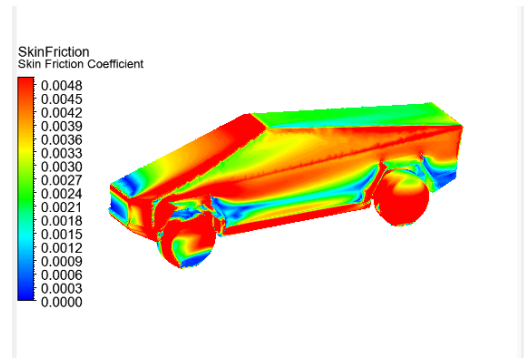


Fig. 9. Skin Friction Coefficient contour for $k - \epsilon$ model

the two turbulence models $k - \epsilon$ and $k - \omega$ SST. In both models, high skin friction coefficients are predicted on the windscreen and roof surfaces, corresponding to the regions of accelerated flows and attached boundary layers, and on the front wheel surfaces exposed to the free stream. Skin friction values on vertical side panels and underbody regions are lower which is in agreement with lower velocity gradients in these regions.

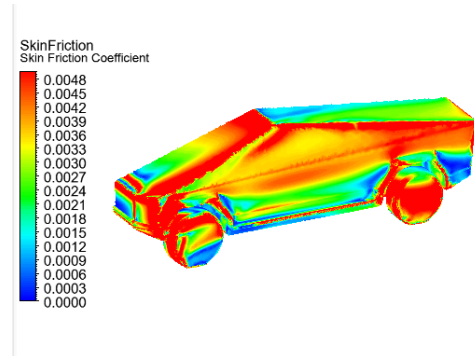
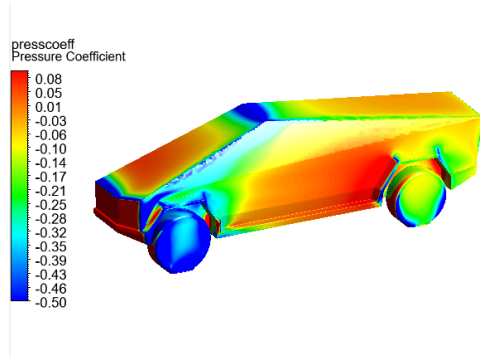


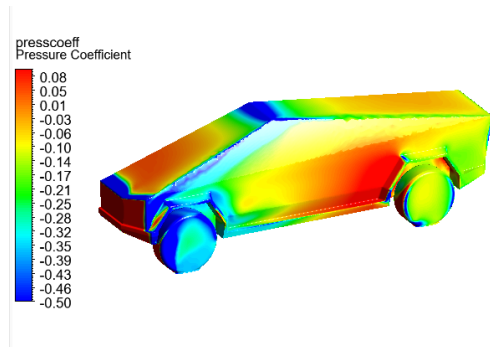
Fig. 10. Skin Friction Coefficient contour for $k - \omega$ model

The $k - \omega$ SST model shows more detailed gradients and sharper changes between high and low friction regions due to its improved near-wall formulation. Significant differences are seen between the models in the near-wall predictions, especially on the bonnet and side surfaces. The $k - \epsilon$ model presents a more smoothed distribution with less variations which is a consequence of its coarser boundary layer resolution. Both models predict low skin friction on the rear surfaces, indicative of flow separation and recirculation, although the spatial extent and magnitude of these low-friction regions are slightly different, which may contribute to the variation in the predicted viscous drag component.

The distribution of pressure coefficient on the truck surface for the $k - \epsilon$ and $k - \omega$ SST turbulence models is shown in Fig.11 and Fig.12. Both formulations predict a modest positive pressure coefficient in the frontal

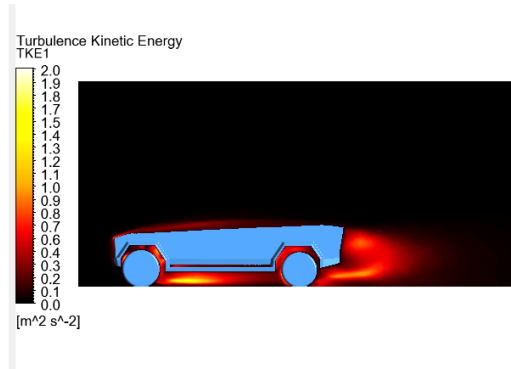
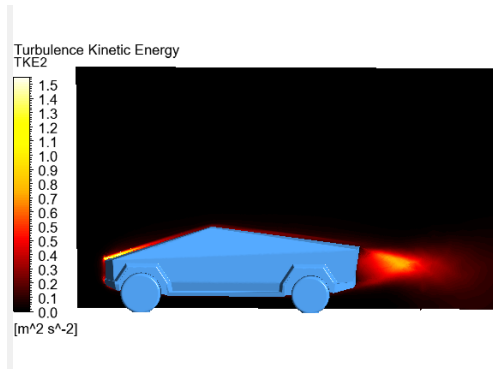
Fig. 11. Pressure Coefficient contour for $k - \epsilon$ model

stagnation region, with values approaching the theoretical maximum, signifying pressure recovery where the freestream flow impacts the vehicle nose. The main part of the vehicle surface, including the roof, bonnet, side panels and the wheel surfaces, is characterised by significant negative pressure coefficients with minimum values of approximately $C_p = -0.50$. The underbody is defined by negative pressures that act to produce aerodynamic lift forces. The pressure coefficient distribution on the rear surface shows strong negative pressure coefficients, which is an indication of the low-pressure wake region generated by flow separation at the blunt trailing edge and is the dominant contributor to pressure drag. The comparison shows that the two turbulence models do not differ significantly in the overall pattern of pressure distribution, predicting similar magnitudes and spatial extents of pressure recovery and suction regions.

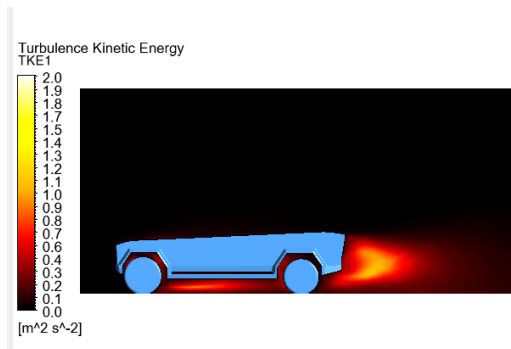
Fig. 12. Pressure Coefficient contour for $k - \omega$ model

Turbulence Model Parameters

The distribution of turbulent kinetic energy on the centerline plane for the $k - \epsilon$ and $k - \omega$ SST turbulence models is shown in Fig.14 and 16. Both contours show that the freestream has negligible turbulence levels and that the attached flow regions on the vehicle surface have low turbulent kinetic energy. The main turbulence production is located in the wake region immediately downstream of

Fig. 13. TKE for $k - \epsilon$ turbulence model on the plane at $z = 0.96$ mFig. 14. TKE for $k - \epsilon$ turbulence model on the centerline plane

the blunt rear surface, where flow separation and strong velocity gradients generate intense mixing and vortical structures. Both models display peaks in the turbulent kinetic energy in the near-wake shear layers and elevated levels downstream as the wake develops and decays. Moderate turbulence production is also observed along the roof edge and the windscreen-bonnet junction, corresponding to the regions of flow acceleration and adverse pressure gradients. Modest levels of turbulent kinetic energy are also observed in the ground boundary layer underneath the vehicle, consistent with wall-bounded shear flow.

Fig. 15. TKE for $k - \omega$ turbulence model on the plane at $z = 0.96$ m

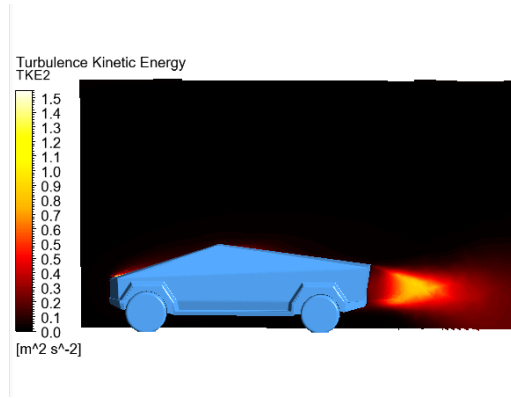


Fig. 16. TKE for $k - \omega$ turbulence model on the centerline plane

A comparative study shows that the two models differ considerably in their predictions of turbulence. The $k - \epsilon$ model shows a broader and more diffuse distribution of wake turbulence with larger values located closer to the rear surface, indicating a rapid generation of turbulence immediately upon separation. Conversely, the $k - \omega$ SST model predicts a more spatially confined turbulence field with sharper gradients at the wake boundaries and slightly lower peak magnitudes, indicating a better resolution of the shear layer development and more physically realistic turbulence transport. The $k - \omega$ model also performs better in terms of turbulence prediction in the underbody region, which is clearly observed in the surface view contours, showing its better near-wall treatment and sensitivity to adverse pressure gradients. Such differences in turbulence structure directly impact the predicted mixing rates, wake recovery, and ultimately the accuracy of drag predictions. The $k - \omega$ SST model generally shows better agreement with experimental observations for separated flows typical of bluff body aerodynamics.

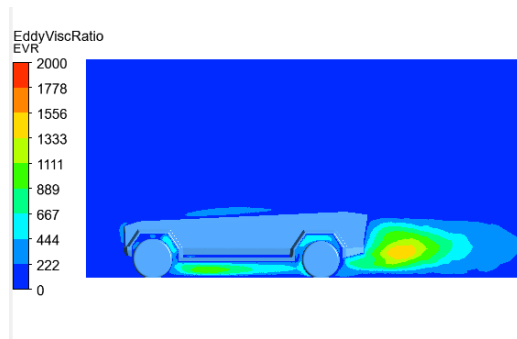


Fig. 17. Eddy Viscosity Ratio contour for $k - \epsilon$ turbulence model on the plane at $z = 0.96$ m

The two models predict low eddy viscosity ratios in the freestream and over most of the vehicle surface where laminar or near-laminar conditions exist. High eddy viscosity ratios are found for the wake region downstream of the blunt rear surface where turbulent mixing and

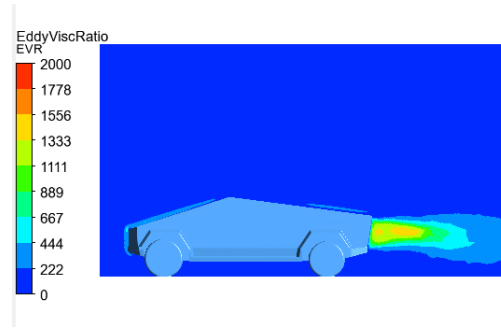


Fig. 18. Eddy Viscosity Ratio contour for $k - \epsilon$ turbulence model on the centerline plane

momentum exchange are strongest due to flow separation and recirculation. Peak values above 1000 are observed in the core of the separated wake, suggesting dominance of turbulent transport over molecular viscosity in these highly disturbed flow regions. The surface contours reveal moderate eddy viscosity ratios in the underbody region and around the wheel wells which are regions of boundary layer development and local flow acceleration.

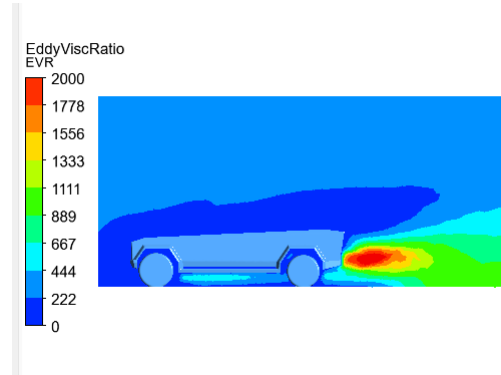


Fig. 19. Eddy Viscosity Ratio contour for $k - \omega$ turbulence model on the plane at $z = 0.96$ m

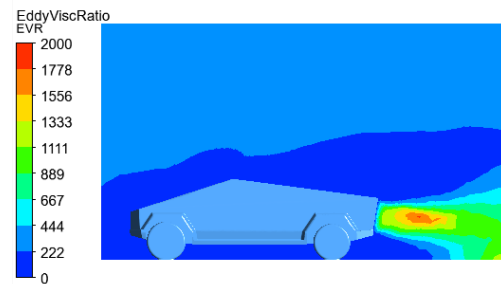


Fig. 20. Eddy Viscosity Ratio contour for $k - \omega$ turbulence model on the centerline plane

The comparison shows that the $k - \epsilon$ model predicts a more spatially diffusive distribution of high eddy viscosity, extending further downstream and laterally

into the wake, consistent with its tendency to overpredict turbulent diffusion in the separated flow regions. On the other hand, the $k - \omega$ SST model provides a more localised distribution of eddy viscosity with steeper gradients at the wake edges and a more compact core of high intensity, which combines the near-wall accuracy of the $k - \omega$ model with the freestream independence of the $k - \epsilon$ approach. These differences in the prediction of eddy viscosity have a direct effect on the calculated turbulent shear stresses, mixing rates, and wake development characteristics, and have implications for the accuracy of drag and flow separation predictions in bluff body aerodynamics.

Author Contribution Statement

The authors confirm contribution to the report as follows: study conception and design: Mukundan Subbaiya(MS), Neha Mandapaka(NM); Setup and simulations: MS, NM; Analysis and interpretation of results: MS, NM; draft report preparation:MS,NM. All authors reviewed the results, approved the final version of the report and declare that each contribution to the work from other peoples published works or unpublished sources have been acknowledged and source of information have been referenced.

Author Declaration of the use of Generative AI and AI-assisted Technologies

During the preparation of this work the authors used Gemini in order to assist with LaTeX formatting, structural development, and technical drafting. Additionally, AI tools were employed to aid in the interpretation and understanding of certain simulation results. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the report content.

REFERENCES

- [1] H. K. Versteeg and W. Malalasekera, *An Introduction to Computational Fluid Dynamics: The Finite Volume Method*. Pearson Education, 2007.
- [2] I. H. Abbott and A. E. von Doenhoff, *Theory of Wing Sections, Including a Summary of Airfoil Data*. New York: Dover Publications, 1959. Standard reference for the NACA 4412 airfoil characteristics.